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Naturwissenschaftlich-Technische Fakultät

4D and Quantum Vision Group

Quantum-enhanced Computer Vision: Going Beyond Classical Algorithms

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Quantum-enhanced Computer Vision: Going Beyond Classical Algorithms

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cs.CV] 8 Oct 2025

Abstract—Quantum-enhanced Computer Vision (QeCV) is a new research field at the intersection of computer vision, optimisation theory, machine learning and quantum computing. It has high potential to transform how visual signals are processed and interpreted with the help of quantum computing that leverages quantum-mechanical effects in computations inaccessible to classical (i.e. non-quantum) computers. In scenarios where existing non-quantum methods cannot find a solution in a reasonable time or compute only approximate solutions, quantum computers can provide, among others, advantages in terms of better time scalability for multiple problem classes. Parametrised quantum circuits can also become, in the long term, a considerable alternative to classical neural networks in computer vision. However, specialised and fundamentally new algorithms must be developed to enable compatibility with quantum hardware and unveil the potential of quantum computational paradigms in computer vision.

This survey contributes to the existing literature on QeCV with a holistic review of this research field. It is designed as a quantum computing reference for the computer vision community, targeting computer vision students, scientists and readers with related backgrounds who want to familiarise themselves with QeCV. We provide a comprehensive introduction to QeCV, its specifics, and methodologies for formulations compatible with quantum hardware and QeCV methods, leveraging two main quantum computational paradigms, i.e. gate-based quantum computing and quantum annealing. We elaborate on the operational principles of quantum computers and the available tools to access, program and simulate them in the context of QeCV. Finally, we review existing quantum computing tools and learning materials and discuss aspects related to publishing and reviewing QeCV papers, open challenges and potential social implications.

Index Terms—Quantum-enhanced Computer Vision, Applied Quantum Computing, Quantum Algorithms.

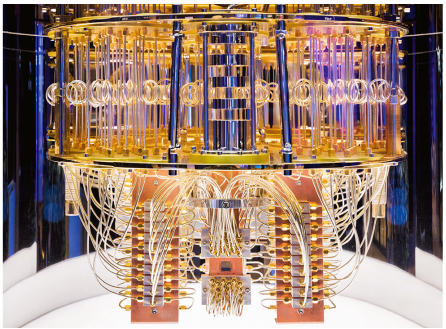
Agenda

1. Background
   - Motivation and Goals
2. Operational Principles
   - Basics and Paradigms
3. Adiabatic Quantum Computing
   - Combinatorial Optimization
4. Gate-based Quantum Computing
   - Generative Models
5. Gate-based Quantum Computing
   - Quantum Interference in Learning

Background

Motivation and Goals

Why Quantum Computing?



Expand computing hardware with an entirely new device type

- CPU/GPU/**QPU**/...
- Compute with laws of **quantum mechanics**.
- **Any computational advantages beyond classical hardware?**

Image source: https://new.abb.com/news/detail/74736/quantum-computing-the-hype-and-hopes

Why Quantum Computing?



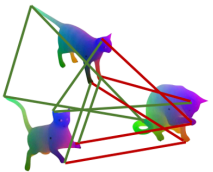
Birdal and Golyanik, 2020

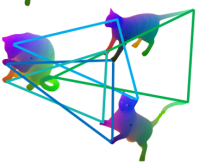


Seelbach Benkner, 2021



Li, 2020

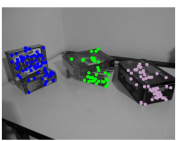




Birdal, 2021



Bhatia, 2023



Farina, 2023



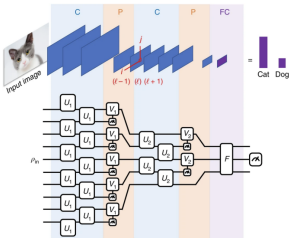
Kuete, 2025

And many more...

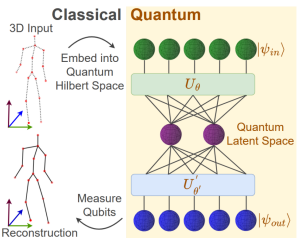
Optimization
& Energy minimization

- Provable quantum speedups exist for specific problems (e.g., Shor/Grover/...).
- **Can similar advantages emerge in computer vision?**

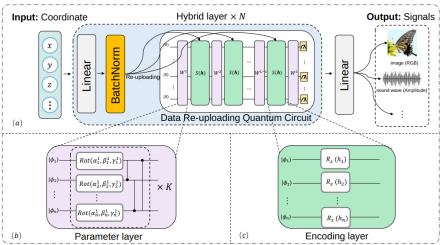
Why Quantum Computing?



Cong, 2019



Rathi, 2023



Zhao, 2024

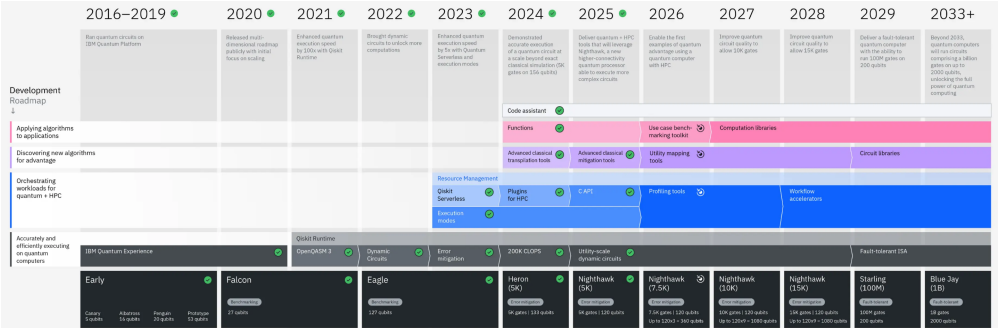
And many more...

Quantum Machine Learning

- We can essentially use and train quantum computers like neural networks.
- **Can quantum actually outperform classical neural networks?**

Quantum Computing Roadmaps

IBM Road Map



And several companies in the race (physical/logical qubit support):

	Year 2026	$\sim$ Year 2032
IBM (Gate model) `https://www.ibm.com/roadmaps/quantum/`	360+/-	- / 2,000+
Google (Gate-based) `https://quantumai.google/roadmap`	100+/-	1M / 10,000+
IonQ (Gate model) `https://www.ionq.com/roadmap`	256+ / 64+	2M / 80,000+
D-Wave (Gate model) `https://www.dwavequantum.com/media/c1wepakh/gate-model-roadmap.jpg`	17/-	- / 100+
D-Wave (Annealer) `https://www.dwavequantum.com/solutions-and-products/product-overview/`	4,400+/-	- / -

IBM roadmap: https://www.ibm.com/quantum/hardware

Operational Principles

Basics and Paradigms

Quantum bit

Any particle that has two basis states and can be in a superposition of them.

Qubit

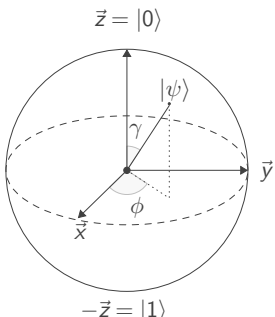
Let $|0\rangle$ and $|1\rangle$ be two basis vectors of $\mathbb{C}^2$, with

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A **qubit** is a one-particle system whose state vector is expressed as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

with $\alpha, \beta \in \mathbb{C}$ subject to $|\alpha|^2 + |\beta|^2 = 1$.



Bloch sphere

Multi-qubit systems

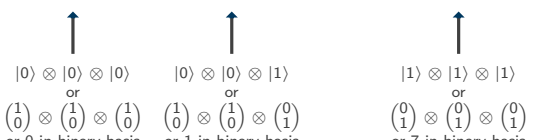
The state of a n-qubit system is given by the **tensor (Kronecker)** product

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle \in \mathbb{C}^{\mathbf{2^n}}$$

of the n single-qubit states.

Basis states are correspondingly formed by tensor products of $|0\rangle$ and $|1\rangle$.
Example for $n = 3$:

$$|\psi\rangle = \alpha_{000}\,|000\rangle + \alpha_{001}\,|001\rangle + \cdots + \alpha_{111}\,|111\rangle$$



Entanglement

Not every multi-qubit states can be written as a tensor product of single-qubit states.

So-called **entangled states** cannot be written in this form.
As an example, for the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \tag{1}$$

there exists no $|\psi_1\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$ and $|\psi_2\rangle = \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$ such that

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \otimes \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1\alpha_2 \\ \alpha_1\beta_2 \\ \beta_1\alpha_2 \\ \beta_1\beta_2 \end{pmatrix}. \tag{2}$$

Measurement in ~~a nutshell~~ the Bloch sphere

Recall

A qubit's state vector

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

collapses to the state $|0\rangle$ with probability to $|\alpha|^2$ and to the state $|1\rangle$ with probability $|\beta|^2$.

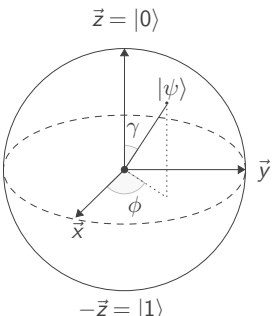
These are merely the results of

$$|\alpha|^2 = \langle\psi|0\rangle\,\langle 0|\psi\rangle \quad \text{and} \quad |\beta|^2 = \langle\psi|1\rangle\,\langle 1|\psi\rangle\,.$$

If we collect $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ in a matrix

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Then $\langle\psi|Z|\psi\rangle = |\alpha|^2 - |\beta|^2$ is the z-coordinate of $|\psi\rangle$ in $\mathbb{R}^3$.



Adiabatic vs. Gate-based QC

At any time $t \in [0, T]$, the evolution of the state vector $|\psi(t)\rangle$ of a quantum system obeys Schrödinger's equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle ,$$

where H is a Hermitian operator known as the system-driven Hamiltonian.

$$i\hbar \tfrac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

$$H(t) = \mathcal{A}(t) H_I + \mathcal{B}(t) H_P$$

Adiabatic quantum computing (AQC)

H_I, H_P: initial and problem Hamiltonians

$\mathcal{A}(t), \mathcal{B}(t)$: scheduling functions

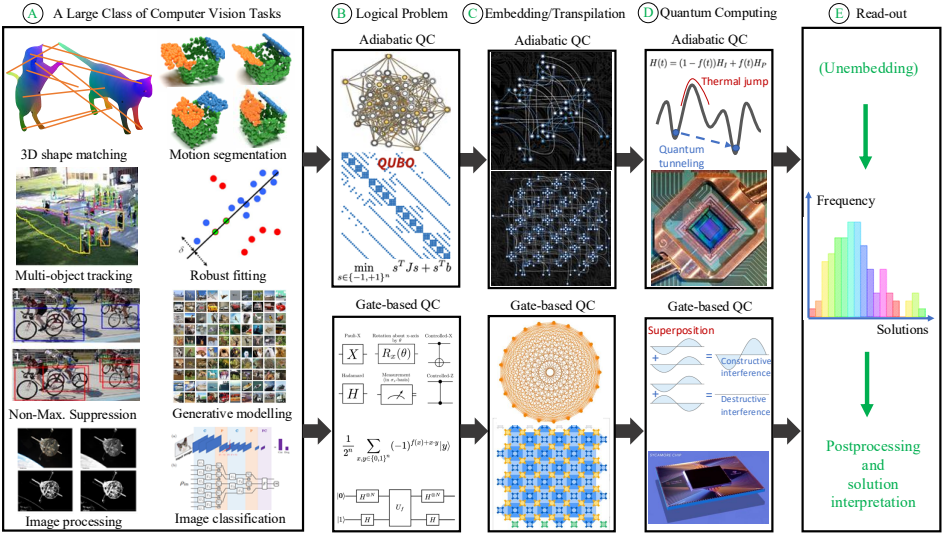
$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Gate-based quantum computing (GQC)

$U(t)$: unitary operator

$U(t)$: depends only on t

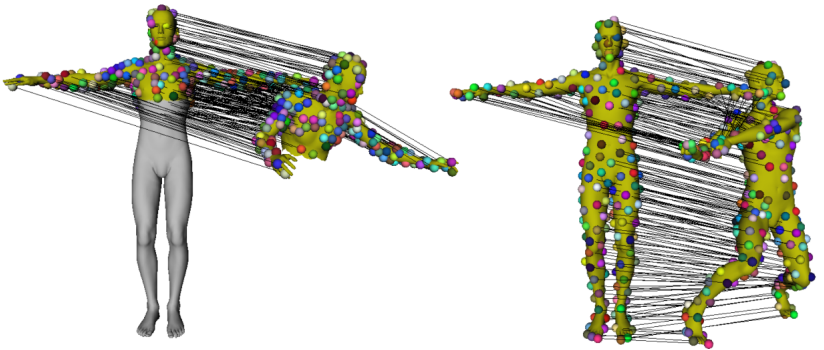
Two Paradigms: AQC vs. GQC



AQC

Combinatorial Optimization

Quadratic Assignment Problems



QAP: Quadratic Assignment Problem

Find a **permutation matrix** P that minimizes

$$\min_{\mathbf{P}\in\{0,1\}^{n\times n}} \sum_{ijkl} \mathcal{D}(d_1(i,j),\, d_2(k,l))\, P_{ik}P_{jl} \qquad \text{s.t.} \qquad \sum_{i=1}^{n} P_{ij} = \sum_{j=1}^{n} P_{ij} = 1.$$

Image source: Ongoing work.

What is AQC Good At?

QUBO: Quadratic Unconstrained Binary Optimization [1]

Given $J \in \mathbb{R}^{k \times k}$, $h \in \mathbb{R}^k$, minimize

$$\min_{s \in \{-1,1\}^k} \quad s^\top J s + h^\top s.$$

Problem Hamiltonian

Define a so-called **problem Hamiltonian**

$$H_P = \sum_{i,j=1}^{k} J_{ij} Z_i Z_j + \sum_{i=1}^{k} h_i Z_i,$$

$$Z_i = \ldots \otimes I \otimes \underbrace{Z}_{\text{pos. } i} \otimes I \otimes \ldots.$$

For a state

$$|q\rangle = \bigotimes_{i=1}^{k} |q_i\rangle\,, \quad |q_i\rangle = \begin{cases} |0\rangle\,, & \text{if} \quad s_i = 1 \\ |1\rangle\,, & \text{if} \quad s_i = -1, \end{cases}$$

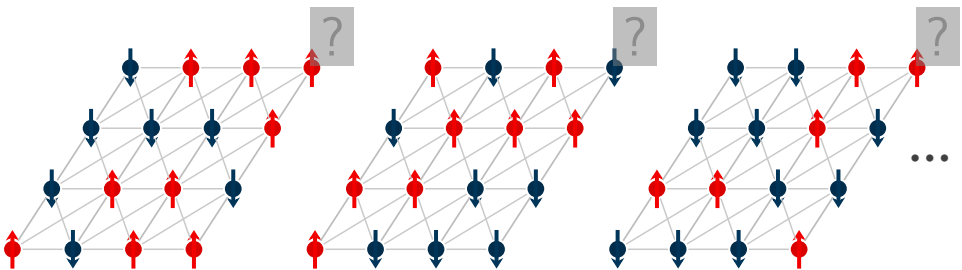
$$\begin{aligned} s^\top J s + h^\top s &= \langle q | H_P | q \rangle \\ &= \text{energy of the quantum system in state } |q\rangle\,. \end{aligned}$$

[1] *A. Das and B. Chakrabarti. Quantum annealing and related optimization methods. Springer Science & Business Media, 2005.*

What is AQC Good At?

So there is an energy associated with each qubit configuration with respect to a given Hamiltonian H_P.

The question becomes: **Which qubit configuration minimizes the energy?**
$\rightarrow$ This would be the minimizer of the QUBO.



What is AQC Good At?

Adiabatic Quantum Computing [1]

A system evolving under the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

and **starting in the ground state of $H(0)$ remains in the ground state of $H(t)$** if the $H(t)$ changes slowly enough and the gap to the first excited state stays sufficiently large.

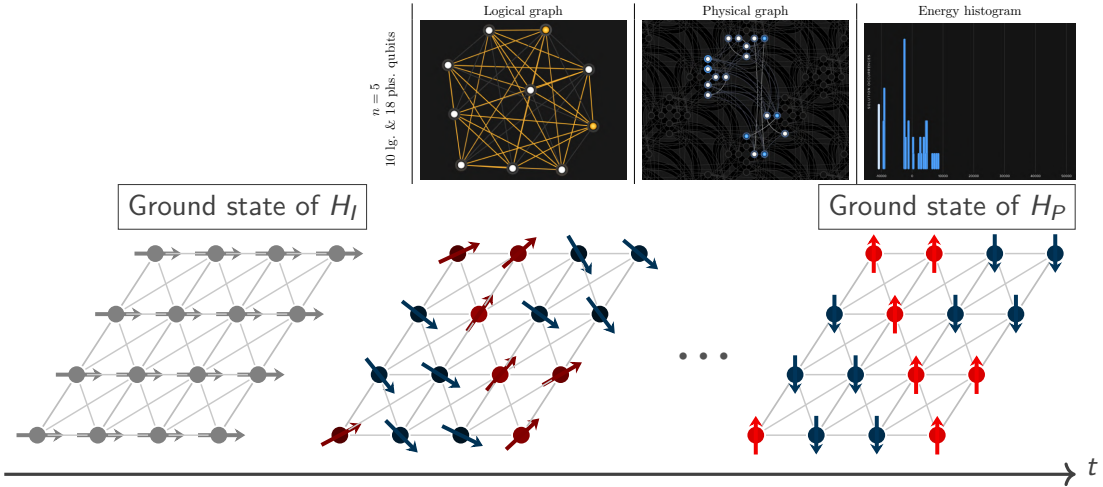
Define an initial H_I whose ground state is easy to prepare and let

$$H(t) = \left(1 - \frac{t}{T}\right) H_I + \frac{t}{T} H_P.$$

The adiabatic theorem brings the system in the ground state of H_P!!! ✔

[1] *T. Albash and D. Lidar. Adiabatic quantum computation. Reviews of Modern Physics, 2018.*

What is AQC Good At?



$$H(t) = \left(1 - \tfrac{t}{T}\right) H_I + \tfrac{t}{T} H_P$$

QAP vs. QUBO

What quantum computers are good at

$$\min_{\mathbf{s}\in\{-1,1\}^k} \quad s^\top Js + h^\top s \quad \overset{\text{Substitution: } s=2x-1}{\longrightarrow} \quad \min_{\mathbf{x}\in\{0,1\}^k} \quad x^\top Qx$$

What we want

$$\min_{\mathbf{P}\in\{0,1\}^{n\times n}} \quad \sum_{ijkl} \mathcal{D}(d_1(i,j),\, d_2(k,l))\; P_{ik}P_{jl} \qquad \text{s.t.} \qquad \sum_{i=1}^{n} P_{ij} = \sum_{j=1}^{n} P_{ij} = 1$$

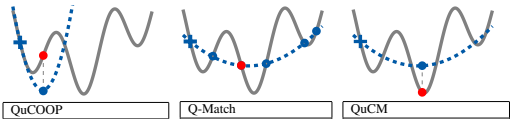
$$\Updownarrow$$

$$\min_{p\in\{0,1\}^{n^2} | Ap=b} \quad p^\top Qp$$

QAP to QUBO

Linearize: $P(x) = \prod_{i=1}^{k} \mathbf{T}_i^{x_i}$

$$= I + \sum_{i=1}^{k} x_i(T_i - I) + \sum_{\substack{i,j=1 \\ i<j}}^{k} x_i x_j (T_i - I)(T_j - I) + \cdots + \prod_{i=1}^{k} x_i (T_I - I)$$



QuCOOP[Kuete Meli et al, 2025]:

$$p(x) = p^{(t)} + \nabla_x p^{(t)} \left(x - x^{(t)} \right)$$

$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x) + \lambda \| p(x) \|_2^2$$

Q-Match[Benkner et al, 2021]:

$$P(x) = \underbrace{\left(\prod_{i=1 \mid T_i \text{ disjoint}}^{k} T_i^{x_i} \right)}_{I + \sum_i x_i (T_i - I)} P^{(t)}$$

$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x)$$

QuCM[under review]:

$$p(x) = \left(I + \sum_i x_i (T_i - I) \right) P^{(t)}$$

$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x) + \lambda \| p(x) \|_2^2$$

QAP Solvers on QAPLIB

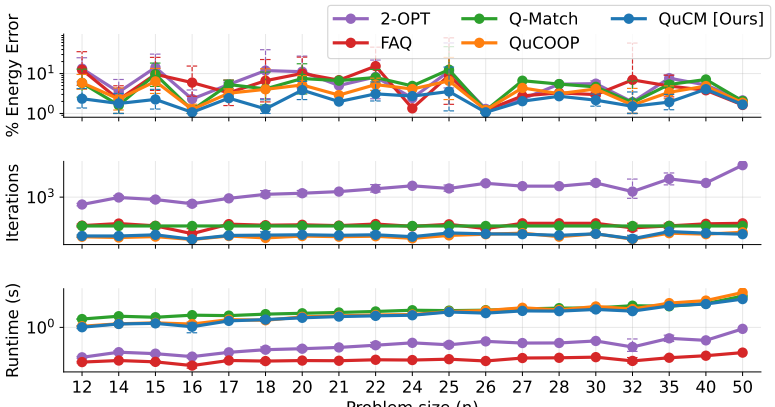


Image source: Ongoing work.

QAP Solvers on Shape Matching

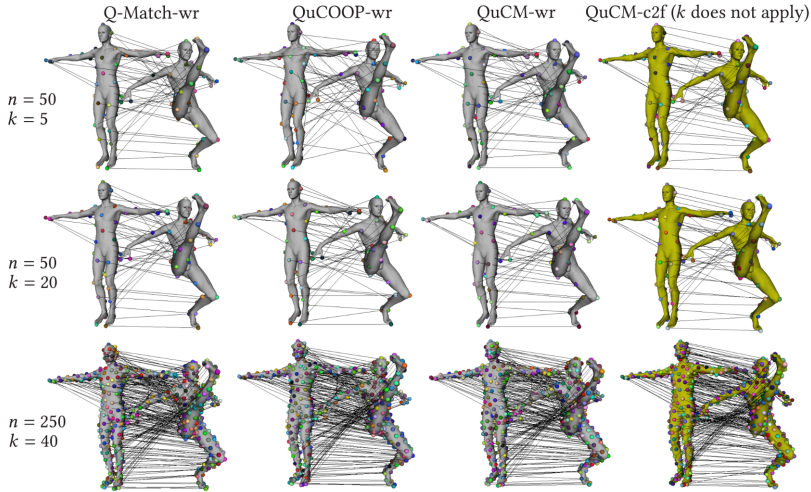


Image source: Ongoing work.

QUBOs and Discrete Optimization

What quantum computers are good at

$$\min_{s\in\{-1,1\}^k} \; s^\top Js + h^\top s \quad \xrightarrow{\text{Substitution: } s=2x-1} \quad \min_{x\in\{0,1\}^k} \; x^\top Qx$$

Quadratic Assignment:

QGM [Seelbach Benkner et al, 2020]

Q-Match[Seelbach Benkner et al, 2021]

QuCOOP[Kuete Meli et al, 2025]

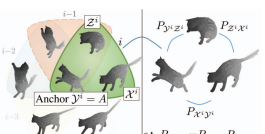


$$\min_{p\in\{0,1\}^{n^2} | Ap=b} \; p^\top Qp$$

Pseudo Boolean Problems:

CCUANTUMM [Bhatia et al, 2023]

QuCOOP[Kuete Meli et al, 2025]



$$\min_{x\in\{0,1\}^n} \sum_{S\subseteq[n]} w_S \prod_{j\in S} x_j$$

Probabilistic Sampling:

K-Means [Zaech et al, 2024]



$p = 0.77$ $p = 0.86$

$$\min_{Z\in\{0,1\}^n} \; -\log p(Z|X)$$

$$p(Z|X) \propto p(X, Z)$$

GQC
Generative Models

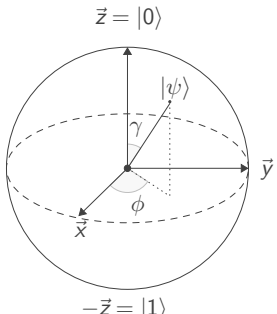
What is GQC Good At?

Computation yields rotating the qubits in the Bloch sphere:

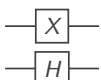
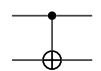
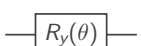
- Requires **unitary transformations** called gates.
- The initial state of the qubits will be assumed to be $|0\rangle$.
- On one qubit, a gate will just be a 2×2 unitary operator

$$U = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

- Take the Kronecker product of single-qubit gates to form a gate on n qubits: $U = U_1 \otimes \cdots \otimes U_n$.
- Use controlled gates to form entangled states.



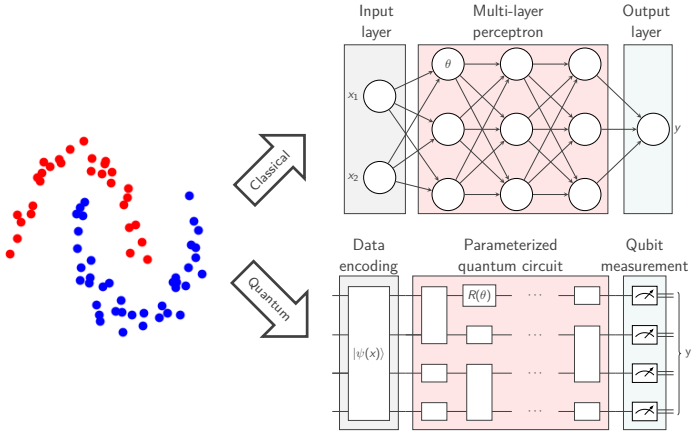
What is GQC Good At?

Gate name	Matrix form	Circuit	Notation
Pauli-X	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$		$X\|0\rangle = \|1\rangle$
Hadamard	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$		$H\|0\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
Multi-qubit gate	$X \otimes H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$		$X \otimes H\|00\rangle = \frac{1}{\sqrt{2}}(\|10\rangle + \|11\rangle)$
Controlled-X on $\|q_0 q_1\rangle$	$I \otimes X^{q_0} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$		$I \otimes X^{q_0}\|00\rangle = \|00\rangle$ $I \otimes X^{q_0}\|10\rangle = \|11\rangle$
Rotation R_y	$R_y(2\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$		$R_y(2\theta)\|0\rangle = \cos(\theta)\|0\rangle - \sin(\theta)\|1\rangle$

Nielsen and Chuang. Quantum Computation and Quantum Information. Cambridge University Press, 2010.
Sutor. Dancing with Qubits: How quantum computing works and how it can change the world. Packt Publishing Ltd, 2019.

What is GQC Good At?

Train circuits like neural networks



Parameter shift-rule [1]:

$$\frac{\partial}{\partial\theta} f = \frac{1}{2}\left(f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})\right)$$

- Differentiable
- Trainable

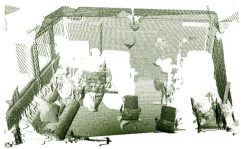
[1] *Schuld et al. Evaluating analytic gradients on quantum hardware. Physical Review A, 2019.*

Data encoding

Consider an input data

$$x = \begin{pmatrix} \vdots \\ x_i \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ 2^0 \cdot q_{x_i}^0 + \cdots + 2^k \cdot q_{x_i}^k \\ \vdots \end{pmatrix} \in \mathbb{R}^d.$$





Basis encoding

$$|\psi(x)\rangle = |q_{x_1}^0 \cdots q_{x_1}^k\rangle \otimes |q_{x_2}^0 \cdots q_{x_2}^k\rangle$$

- Requires dk qubits.

Angle encoding

$$|\psi(x)\rangle = R_x(x_1)\,|0\rangle \otimes R_x(x_2)\,|1\rangle$$

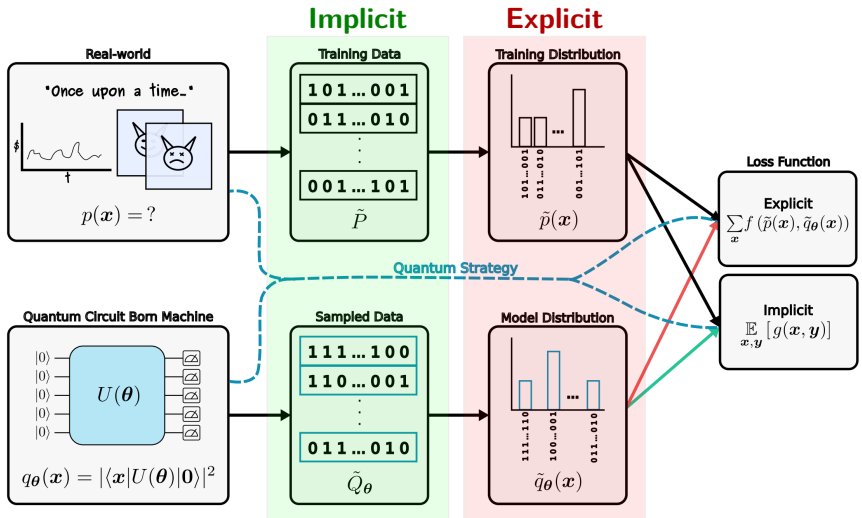
- Requires d qubits.

Amplitude encoding

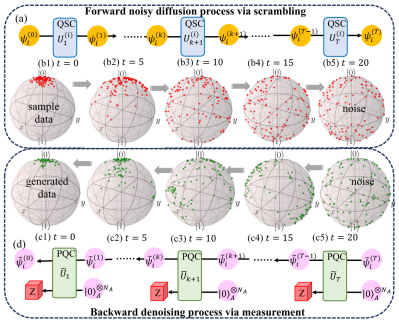
$$|\psi(x)\rangle = \frac{1}{\|x\|}(x_1\,|0\rangle + x_2\,|1\rangle)$$

- Requires $\log_2(d)$ qubits.

Quantum Generative Models: Implicit vs. Explicit

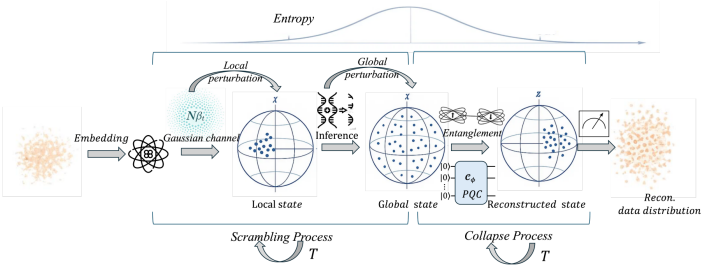


Quantum Diffusion Models



Quantum Denoising Diffusion Probabilistic Models (QDDPM):

- Hard to sample $x_T \sim p_T(x)$

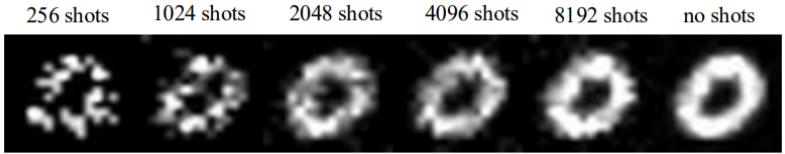


QDDPM with Gaussian Channels (QGen:)

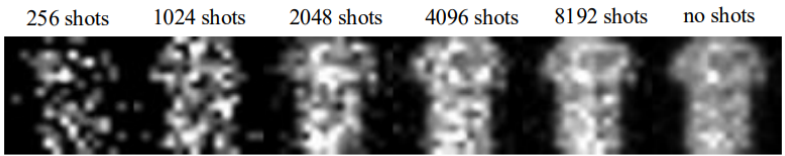
- Gaussian noise for local perturbation
- Unitary scrambling for global perturbation

QDDPM: Zhang et al. Generative Quantum Machine Learning via Denoising Diffusion Probabilistic Models. Physical Review Letters 2024.
QGen: Li et al. A purely Quantum Generative Modeling through Unitary Scrambling and Collapse. ICLR 2026.

Quantum Diffusion Models



(a) MNIST



(b) FashionMNIST

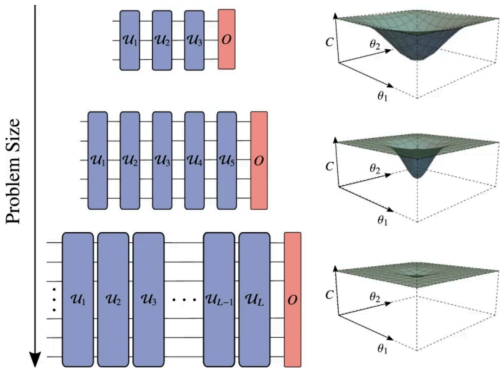
Image generation with QGen.

Trainabity Challenges

- **Barren Plateau[1]:**

  "The gradient vanishes exponentially in the number of qubits n if the depth of the ansatz grows linearly with n."

- **Parameter shift rule[2]:**
  $\frac{\partial}{\partial\theta} f = \frac{1}{2}\left(f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})\right)$.

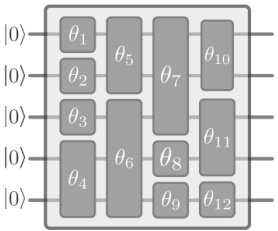


Sampling noise + parameter shift rules $\Longrightarrow$ Prohibitive sampling overhead!

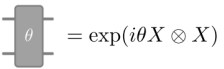
$\rightarrow$ **We need to design around trainability issues...**

[1] *Wang et al. Noise-induced barren plateaus in variational quantum algorithms. Nature Communications, 2021.*
[2] *Schuld et al. Evaluating analytic gradients on quantum hardware. Physical Review A, 2019.*

Quantum Implicit Generative Models with IQPs



Instantaneous Quantum Polynomial (IQP) circuits



Training with MMD^2 loss

For a feature map ϕ and Gaussian kernel k:

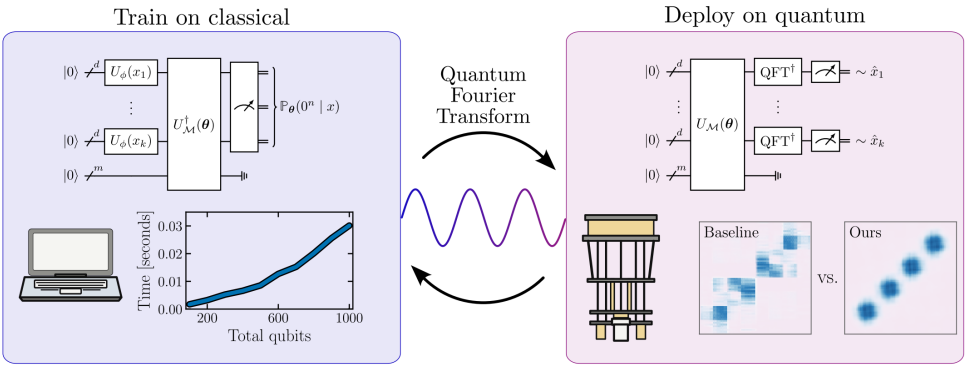
$$\begin{aligned}\mathrm{MMD}^2(p, q_\theta) &= \|\mathbb{E}_{x\sim p}[\phi(x)] - \mathbb{E}_{y\sim q\theta}[\phi(y)]\|^2 \\ &= \mathbb{E}_{x,y\sim p}[k(x,y)] \\ &\quad - 2\mathbb{E}_{x\sim p, y\sim q_\theta}[k(x,y)] \\ &\quad + \mathbb{E}_{x,y\sim q\theta}[k(x,y)]\end{aligned}$$

It turns out that for IQP circuits (after some Hadamard sandwichs)

$$\begin{aligned}\mathrm{MMD}^2(p, q_\theta) &= \mathbb{E}_{a\sim\mathcal{P}_\sigma(a)}[(\langle Z_a\rangle_p - \langle Z_a\rangle_{q_\theta})^2] \\ \langle Z_a\rangle_p &= \mathbb{E}_{x\sim p}[(-1)^{x\cdot a}] \\ \langle Z_a\rangle_{q_\theta} &= \mathbb{E}_{z\sim\mathcal{U}}[\cos\big(\sum_j \theta_j (-1)^{g_j\cdot z}(1-(-1)^{g_j\cdot a})\big)]\end{aligned}$$

Train on Classical → Deploy on Quantum

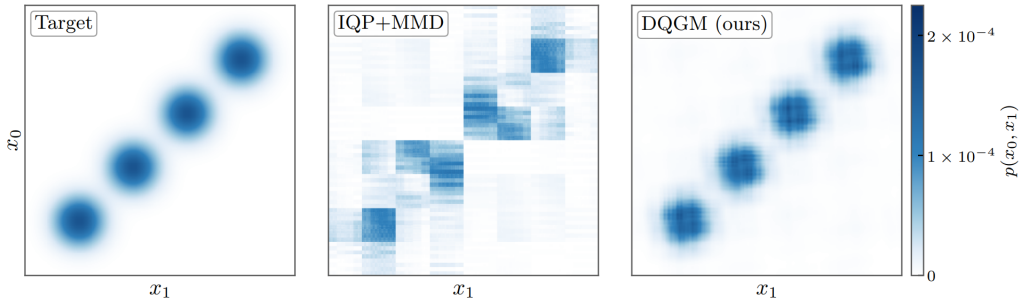
Implicit Quantum Generative Models with DQGM



Training with negative log-likelihood (NNL) loss

$$\mathrm{NNL}(\theta) = -\mathbb{E}_{x\sim p}[\log P_\theta(0^n|x)], \quad P_\theta(0^n|x) = \mathbb{E}_{\omega\sim\mathcal{U}}[|\prod_{v=1}^{n}\frac{1}{2}(r_v(0;\omega)+r_v(1;\omega))|^2]$$

Implicit Quantum Generative Models



GQC

Quantum Interference in Learning

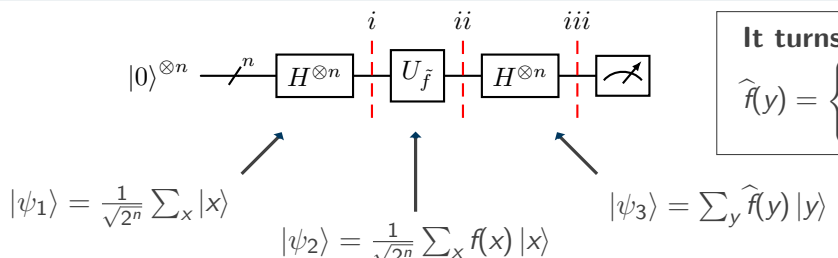
What are Quantum Computers Good At?

Bernstein-Vazirani [1]

Given observations $(x, f(x))$ with the assumption

$$f(x) = s^\top x \bmod 2,$$

for $x, s \in \{0,1\}^n$, recover the *unknown* s.



It turns out that:

$$\widehat{f}(y) = \begin{cases} 1, & \text{if } y = s, \\ 0, & \text{otherwise.} \end{cases}$$

[1] *E. Bernstein and U. Vazirani. Quantum complexity theory. ACM symposium on Theory of computing, 1993.*

Bernstein-Vazirani vs. Learning

What quantum computers are good at

Given observations $(x, f(x))$ with the assumption

$$f(x) = s^\top x \bmod 2,$$

for $x, s \in \{0,1\}^n$, recover the *unknown* s.

What we want

Function f does not necessarily fulfill

$$f(x) = s^\top x \bmod 2.$$

- $(x, f(x))$: Training samples
- $f(x) = s^\top x \bmod 2$: Model, architecture
- s: Model parameters

What are Quantum Computers Good At?

Theorem (Fourier expansion on $\mathbb{Z}_2^n$, [O'Donnell, 2014])

Any function $f\colon \{0,1\}^n \to \mathbb{R}$ admits a unique expansion

$$f(x) = \sum_{y\in\{0,1\}^n} \widehat{f}(y)\,\chi_y(x), \quad \chi_y(x) = (-1)^{y^\top x}$$

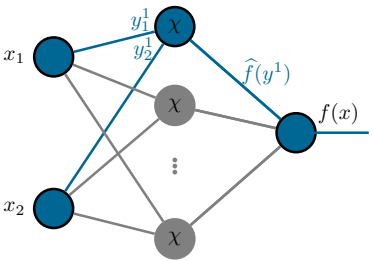
called the Fourier expansion of f, and $\widehat{f}(y) \in \mathbb{R}$ are Fourier coefficients in the basis χ.

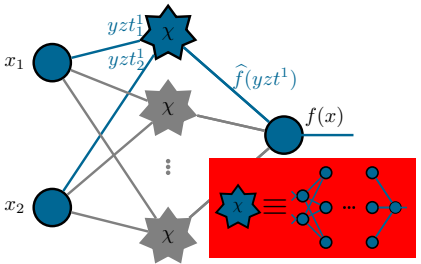
From $|\psi_3\rangle = \sum_y \widehat{f}(y)\,|y\rangle$:

- Sample y with prob. $|\widehat{f}(y)|^2$
- Knowing the expression of χ, recover $\widehat{f}(y) = \sum_x f(x)\chi_y(x)$ from training data.
- **Efficient approximation:** $f(x) = \sum_{y \text{ sampled}} \widehat{f}(y)\chi_y(x)$.

[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

Bernstein Vazirani Networks (BVNs)





Standard:

$$f(x) = \sum_y \widehat{f}(y)\,\chi_y(x)$$

$$\chi_y(x) = (-1)^{y^\top x}$$

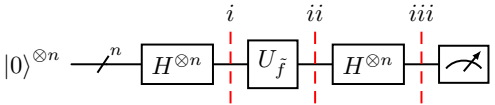
Generalized:

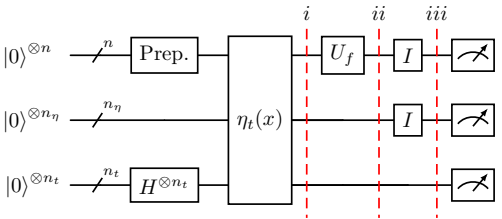
$$f(x) \approx \sum_{yzt} \widehat{f}(y)\,\chi_{yzt}(x)$$

$\chi_{yzt}(x) =$ **Something more expressive**

[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

Bernstein Vazirani Networks (BVNs)





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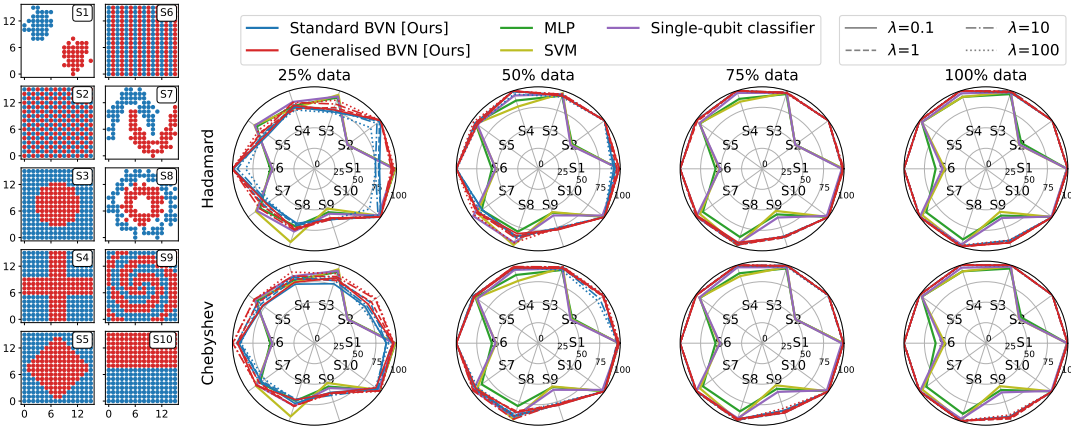
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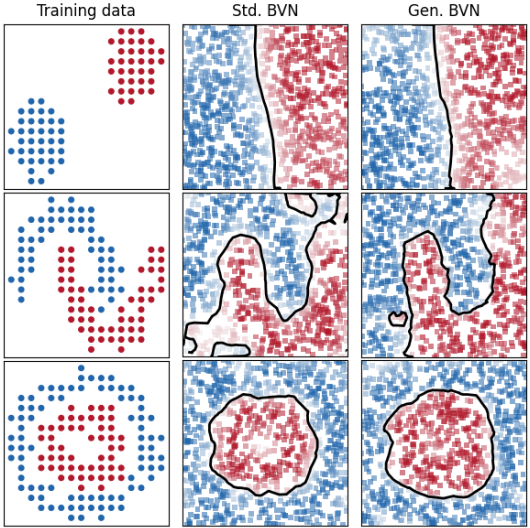
[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

BVNs on Binary Classification



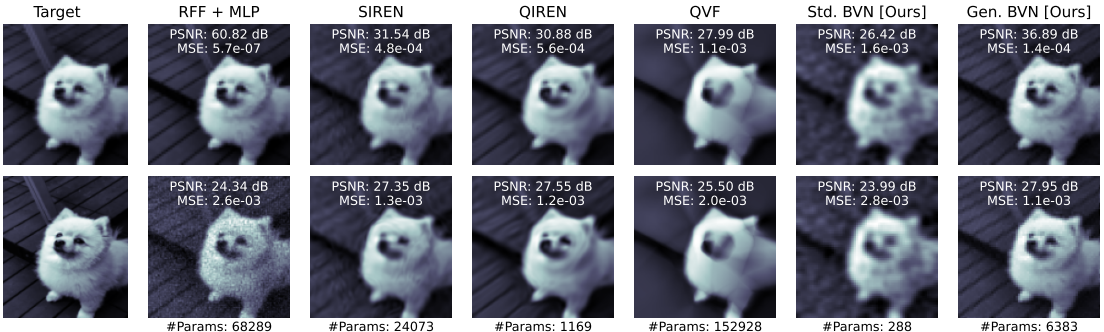
[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

BVNs on Binary Classification



[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

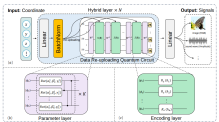
BVNs on Implicit Image Representation



[1] *Kuete Meli et al. Bernstein-Vazirani Networks: Quantum Machine Learning by Interference, ArXiv, 2026.*

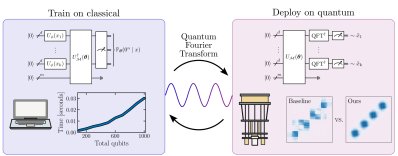
Summary





Adiabatic vs. Gate-based QC:

- AQC is a good heuristic for optimization.
  Map CV problems to QUBOs first.
  Prefer compact encodings, exploit sparsity.
- GQC offers flexibility in algorithm design.
  Analytical is great but not obvious; training is an option.
  Design around barren plateaus and sampling overhead.



Several emerging QeCV directions:

- Quantum computers are natural samplers.
- Quantum enables expressive spectral models.
- Exploit quantum–classical synergy.

Thanks! Questions?

natacha.kuetemeli@uni-siegen.de